FRONT PAGE OF THE ANSWER SCRIPT

- Examination (Write the official title of the examination as it appears at the head of the question paper: ……………………………………………… ……………………………………………… ………………………………………………

- Title of the paper and Subject code: ……………………………………………… ………………………………………………

- Index Number (Write very clearly): ……………………………………… ………………………………………

EXAMPLE

- Examination (Write the official title of the examination as it appears at the head of the question paper: BSc Second Year Semester II Examination

- Title of the paper and Subject code: Organic Chemistry II, CHE 2202

- Index Number (Write very clearly): 0000

- Registration Number: AsB/2017/2018/000

- **Use the following format to name your answer script:**
  CHE 2202_Index#_Registration#

  Eg: CHE 2202_4000_ASB 2017.2018.000

- **Upload the answer script to LMS or email to :- che2202@as.rjt.ac.lk**

- **Any inquiry call / whatsapp /viber : 0712943957 / 0715408587 / 0716317300**

RAJARATA UNIVERSITY OF SRI LANKA
FACULTY OF APPLIED SCIENCES
B.Sc. (General) Degree in Applied Sciences
Second Year Semester I Examination – January / February 2021

CHE 2202 – ORGANIC CHEMISTRY II

Answer all four questions. **Time: 02 hours**

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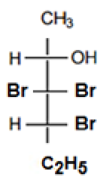
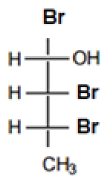
1. a). Name the following compounds using RS or EZ nomenclature. Draw all the necessary steps and write IUPAC names of each compound.

i).



ii).



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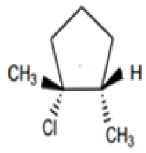

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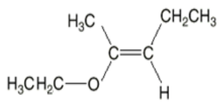
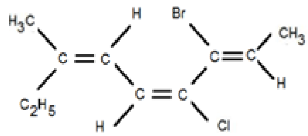


v).



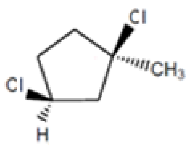
(10 marks)

b). i) Draw the Fischer and Newman projections of (2R,3R,4R)-3,4-Dibromo-2-methylhexane (03 marks)

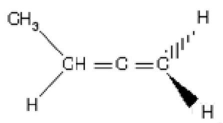
ii) Draw the structure of (3Z,5E)-3-Chloro -4,5-dimethyl-3,5-octadiene (02 marks)

c) Define "Stereoisomers" and state whether the following compounds are optically active or inactive. Explain your answer.

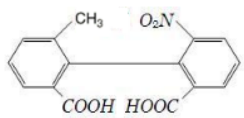
i)



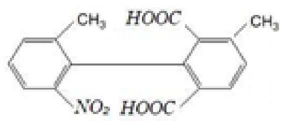
ii)



iii)



iv)



(10 marks)

2. a) In the structure of Chlorocylclohexane, equatorial chloro group is “anti” to C3 in the ring, while axial chloro group is “gauche” to C3 in the ring. Draw chair conformations and Newman projections and explain the statement.

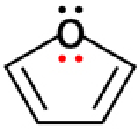
(06 marks)

b). Draw the conformations of 1,2 dibromoethane using Newman projection formulae and plot the potential energy vs angle of rotation curve for the rotation of C1 – C2 bond through 360°C

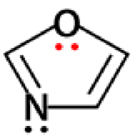
(06 marks)

c). i. Discribe Huckel’s rule for Aromaticity (03 marks)

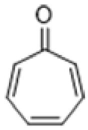
ii. Determine whether the following compounds show aromaticity according to Huckels rule. Explain your answer



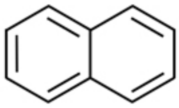
(i)



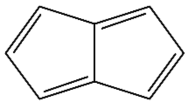
(ii)



(iii)



(iv)

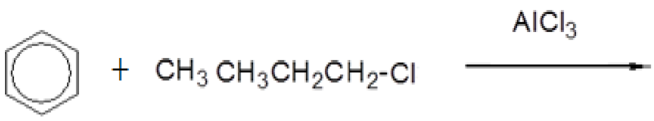


(v)

(10 marks)

3 a). Write the product/s and complete the mechanisms of the following reactions. .

i.



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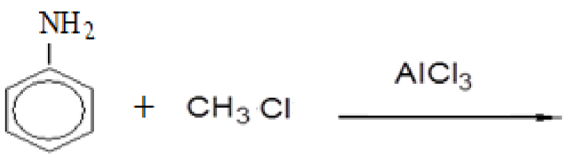
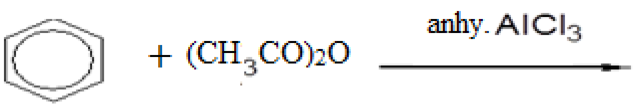


iii.



(06 marks)

b). laboratory preparation of phenols involves melting aromatic sulfonic acids with NaOH at high temperature. Write down the detailed three- step mechanism for the synthesis of p-ethylphenol from benzene.

(05 marks)

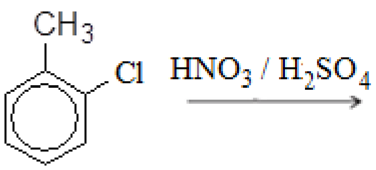
c). Substituted phenols can be more or less acidic than phenol itself. Explain the statement with a detailed mechanism.

(06 marks)

d). Discuss a mechanism to systhesis of p-nitropropyl benzene starting from benzene.

(05 marks)

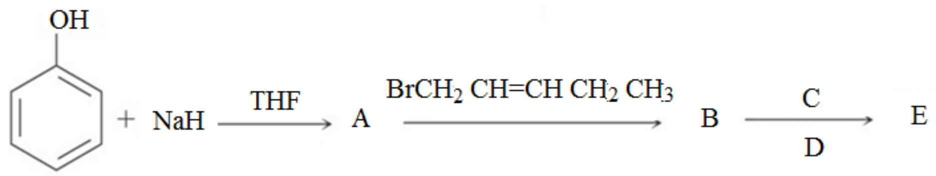
e). Discuss the products of the following reaction.



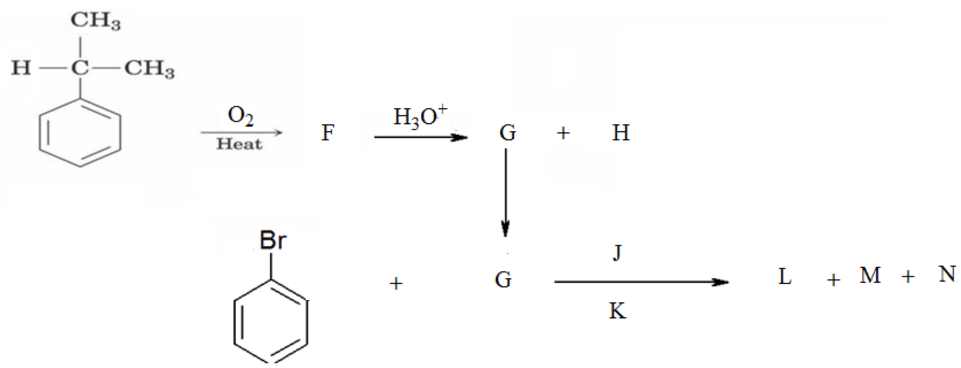
(03 marks)

4. a). Identify unknown compounds A, B, C, D, E, F, G, H, K, L, M and N, and complete the following reactions. Write the mechanism for the conversion of B into E.

i.

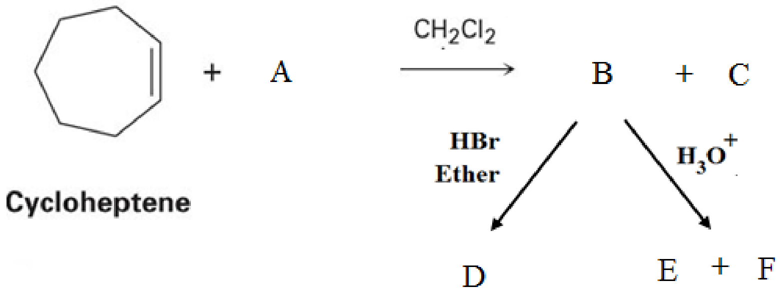


ii.



(10 marks)

b). Identify unknown compounds A, B, C, D, E and F, and complete following the reactions



(06 marks)

c) Write a a four-step reaction mechanism to prepare phenyl acetic acid from benzene.

(04 marks)

d) Write a short account on "vat dyes" and "disperse dyes" used to color fabricks. Draw the structures of the corresponding dyes.

(05 marks)

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